22OE1ME303

R22

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)

IV B.Tech., II Semester, Sessional Examinations - I, February 2026

3D PRINTING AND DESIGN
(OPEN ELECTIVE (General Track))

Time: 120 Min. Date of Examination: 25-02-2026 Max. Marks: 30M

Draw diagrams wherever necessary

PART-A

I. Answer the following. **5X2=10M**

1. Explain the differences between Conventional Manufacturing and Additive Manufacturing processes. (CO-1, BL-2)
2. List the various real-world applications of 3D printing technology. (CO-1, BL-2)
3. Define photopolymerization. (CO-2, BL-1)
4. Discuss the advantages and disadvantages of the Material Jetting process. (CO-2, BL-1)
5. List the key characteristics of the Fused Deposition Modeling (FDM) process. (CO-3, BL-1)

PART-B

II. Answer one question from each unit. **20 M**

UNIT-I

1. Explain the complete 3D printing process chain in detail. (CO-1, BL-2) [8M]

(OR)

2. Explain the differences between 3D Printing and Conventional Manufacturing. (CO-1, BL-2) [8M]

UNIT-II

3. Explain the principle and working process of Stereolithography (SLA) with a neat sketch. (CO-2, BL-3) [8M]

(OR)

4. Describe the principle and working process of Material Jetting with a neat sketch. (CO-2, BL-4) [8M]

UNIT-III

5. Explain the Fused Deposition Modeling (FDM) process in detail with a neat sketch. (CO-2, BL-3) [4M]

(OR)

6. Discuss the advantages and disadvantages of the Fused Deposition Modeling (FDM) process. (CO-2, BL-2) [4M]

****

22071A6732

22PE1DS403 | R22

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

IV Year B.Tech II Semester Sessional Examinations-1, February,2026

ADVANCED DATA SCIENCE

(CSE-DS and AI&DS)

Time: 120 Mins | DOE: 23.02.2026 (AN) | Max. Marks: **30 M**

PART-A

Answer ALL questions **5 X 2 = 10M**

1. Define the role of the Learning Rate in the Gradient Descent algorithm (CO-1) [BL-2]
2. What is the primary difference between Batch Gradient Descent and SGD? (CO-1) [BL-2]
3. Explain the significance of 'Padding' and 'Strides' in a Convolutional layer. (CO-2) [BL-2]
4. Define Max-pooling and its impact on the spatial dimensions of an image. (CO-2) [BL-2]
5. What is a 'Receptive Field' in the context of Convolutional Neural Networks? (CO-3) [BL-1]

PART-B

Answer any ONE from each group **2 X 8 = 16M**
1 X 4 = 4M

UNIT-I

1. Explain the mathematical formulation of Logistic Regression and describe how Gradient Descent is used to update weights. [8M](CO-1)[BL-3]

OR

2. Describe the process of Principal Component Analysis (PCA) as a constrained optimization problem for dimensionality reduction. [8M](CO-1)[BL- 3]

UNIT-II

3. Explain the architecture of a standard Convolutional layer and RGB processing. [8M] (CO-2)[BL-3]

OR

4. Explain common optimization techniques used in CNN training. How does data augmentation help reduce overfitting. [8M] (CO-2)[BL- 3]

UNIT-III

5. Explain Receptive Field and Effective Receptive Field in CNN. [4M] (CO-3)[BL-2]

OR

6. Discuss the architecture of LeNet and its significance. [4M] (CO-3)[BL-2]

22PE1DS404

R22

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

IV Year B.Tech II Semester Sessional Examinations-1, February, 2026

GENERATIVE AI ESSENTIALS

(CSE-DataScience)

Time: 120 Mins | DOE: 24.02.26 (A.N) | Max. Marks: 30 M

PART-A

Answer ALL questions 5 X 2 = 10M

1. Define Artificial Intelligence and mention two applications. (CO-1) [BL-1]
2. Differentiate between parameters and hyperparameters (CO-1) [BL-2]
3. For input $x = 3$, weight $w = 0.4$, bias $b = 0.2$, find neuron output using linear activation. (CO-2) [BL-2]
4. What is the role of forget gate in LSTM? (CO-2) [BL-1]
5. What is temperature parameter in text generation? (CO-3) [BL-1]

PART-B

Answer any ONE from each group 8 X 2 = 16M

4 X 1 = 4M

UNIT-I

1. Explain the history and evolution of AI and the Deep Learning revolution. [8M] (CO-1)[BL-2]

OR

2. Describe the structure of Artificial Neural Networks and explain the steps involved in training with mathematical expressions. [8M] (CO-1)[BL-4]

UNIT-II

3. Explain Long Short-Term Memory (LSTM) architecture with neat diagram and equations [8M] (CO-2)[BL-2]

OR

4. Explain the working principle of GAN. Why is it considered a minimax optimization problem?. [8M] (CO-2)[BL-2]

UNIT-III

5. Describe Transformer architecture. Explain Self-Attention mechanism with formula and advantages over RNN. [4M](CO-3)[BL-2]

OR

6. Discuss text generation using LSTM and Transformer models. Explain fine-tuning of language models. [4M] (CO-3)[BL-4]

----------XXXX----------